

AHSEN TAHIR

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Education

B.S. in Computer Science, CGPA 3.55 / 4.00

Aug 2023 – May 2027 (expected)

National University of Computer and Emerging Sciences (FAST NUCES), Karachi, Pakistan

- Final-year thesis: detecting whether open-weight LLMs were trained on reasoning traces distilled from frontier models.
- Dean's List: Fall 2023, Spring 2024, Fall 2024.

Aspire Leaders Program

Jan – Apr 2024

Harvard Business School (remote)

Research Interests

LLM post-training: instruction tuning, preference optimization, knowledge distillation. **AI safety:** keeping models faithful to user intent under untrusted input, in particular prompt-injection defenses trained into the weights. **Agentic AI:** coordination and communication among autonomous LLM agents. **Evaluation:** long-horizon agent tasks, and the security of the code agents generate.

Publications

Anemoi: A Semi-Centralized Multi-agent System Based on Agent-to-Agent Communication MCP Server from Coral Protocol

Xinxing Ren, Caelum Forder, Qianbo Zang, Ahsen Tahir, Roman J. Georgio, Suman Deb, Peter Carroll, Önder Gürçan, Zekun Guo

NeurIPS 2025 Workshop on Language Agents and World Models (LAW) · [arXiv:2508.17068](#)

- Replaces a single all-knowing planner with structured agent-to-agent communication over an MCP server: worker agents monitor and refine the plan during execution, instead of following one dictated up front by a planner that cannot see execution state.
- 52.73%** on the GAIA benchmark with a deliberately small planner model (GPT-4.1-mini), **+9.09%** over the OWL baseline under identical LLM settings, with the largest gains on long-horizon tasks where centralized planning degrades most.
- Key finding: the coordination structure, not planner size, was the binding constraint; distributing the planning burden let a smaller model outperform a centralized system built on a larger one.

Beyond Rule-Based Workflows: An Information-Flow-Orchestrated Multi-Agents Paradigm via Agent-to-Agent Communication from CORAL

Xinxing Ren, Qianbo Zang, Caelum Forder, Suman Deb, Ahsen Tahir, Roman J. Georgio, Peter Carroll, Zekun Guo

arXiv preprint, January 2026 · [arXiv:2601.09883](#)

- Drops the fixed workflow graph entirely: agents communicate in natural language over an A2A protocol while a centralized information-flow orchestrator monitors task state and adapts coordination dynamically, rather than executing a predetermined graph.
- 63.64%** pass@1 on GAIA, **+8.49%** over the workflow-based OWL baseline at comparable token usage, with notably better behavior on edge cases and long-horizon task decomposition.

Research Experience

Mitacs Globalink Research Intern

Jun 2026 – Present

University of New Brunswick, Fredericton, Canada · with Prof. Francis Palma

- First author on *When Agents Architect: Design-Level Security Anti-Patterns in LLM-Generated Microservice Systems* (with F. Palma, S. Khan), in progress: whether coding agents introduce security flaws at the architectural rather than the line level.
- Operationalized a catalog of design-level anti-patterns from the microservice security-smells literature, a class of flaw committed when the architecture is decided rather than when a line is written, which line-level CWE analysis does not measure.
- Built the agentic detection pipeline: indexes a microservice repository, reconstructs its structural views, and returns a three-valued verdict per anti-pattern, where static analysis misses the class entirely.

SASD: Source-Aware Self-Distillation

2026 – Present

Collaborative research project · targeting NeurIPS / ICLR

- Training-time defense against prompt injection. An agent reads retrieved pages, tool output, and its operator's instructions through one undifferentiated channel; existing defenses patch this at inference and fall to an attacker who rewords the payload.
- Method: the student is optimized against a multi-teacher KL divergence objective where each teacher sees the same task under a different source condition, targeting the teacher that never saw the injected span, so source tracking ends up in the weights rather than policed by a wrapper at run time.
- Building the training stack and injection benchmark suite; evaluating against the strongest current defenses on attack templates held out at training, including adaptive attacks that optimize against the defense itself.

AI Engineer

Feb 2025 – Jul 2026

CoralOS, London, UK (remote)

- Built agent-to-agent communication infrastructure used in production multi-agent systems, including structured protocols for task orchestration.
- Evaluated interoperability across multiple LLM agent frameworks: prompt design, dynamic role delegation, and structured output pipelines.
- This work led directly to the two publications above.

Teaching & Outreach

Demystifying the Mitacs Globalink Research Internship

2026

Public walkthrough · [YouTube](#)

- A 20-minute walkthrough of the Mitacs Globalink application for undergraduates outside Canada: eligibility, how the timeline actually runs, how to approach professors, and what a competitive application looks like.
- Made after going through the process myself, because available guidance was thin and mostly wrong about the parts that matter; students from countries with little institutional support for research applications lose most to that information gap.

Other Experience

Co-Founder

Feb 2026 – Present

Velton, Fredericton, Canada

- Co-founded the company and led its pivot from a services agency to a product-led startup building OnClickAI, an embedded copilot layer for B2B SaaS products.

Deputy Head, Automation Team

Jan – Apr 2025

PROCOM, Karachi, Pakistan

- Built a WhatsApp chatbot handling student queries for PROCOM’25, with context retention across turns, FAQ resolution, PDF delivery, and fallback to a human representative; added real-time intervention for organizers and query logging for analytics.

Content Writer (freelance)

Dec 2023 – Jul 2024

Estrakt, Remote

- Wrote long-form articles explaining technical developments for a general audience, technical writing practice that carries directly into papers and research communication.

Technical Skills

LLM post-training: supervised fine-tuning, instruction tuning, preference optimization (DPO, RLHF), knowledge distillation and multi-teacher objectives, KL-divergence training objectives, parameter-efficient fine-tuning (LoRA, QLoRA), quantization, synthetic training-data generation.

AI safety & robustness: prompt-injection defense, adversarial and adaptive attack evaluation, security-utility trade-offs, secure-code evaluation.

Agentic AI: multi-agent orchestration, agent-to-agent (A2A) protocols, Model Context Protocol, LangGraph, LangChain, CrewAI, tool use, long-horizon task decomposition.

Evaluation: benchmark design, held-out and adaptive evaluation protocols, ablation design, LLM evaluation.

ML / DL: PyTorch, TensorFlow, JAX, Hugging Face, attention architectures, RAG, vector databases (Pinecone, Qdrant, FAISS), computer vision (YOLO, OpenCV).

Engineering: Python, C/C++, SQL, TypeScript, FastAPI, Next.js, Docker, Git.

Selected Projects

Applied ML systems built end to end; source and detail at [ahsentahir.github.io/projects](#).

- **Trend-X-BTC:** 30-day Bitcoin forecasting with stacked LSTMs feeding 4-head self-attention, trained over four aligned signal families: market data, news sentiment, on-chain activity, and macro series from FRED.
- **Eeko AI:** agriculture assistant for smallholder farmers; Llama 3.2 Vision for crop pest and disease detection under task-specific prompts, grounded in NASA LARC satellite data for the user’s region, with Urdu speech support for accessibility.
- **VisionPlay:** player and ball tracking in broadcast football footage with fine-tuned YOLO and ByteTrack; camera motion estimated by Lucas-Kanade optical flow and removed before a perspective homography maps tracks onto pitch coordinates.

Awards

- **Mitacs Globalink Research Internship**, 2026: fully funded international research placement in Canada, awarded through multi-stage screening and faculty matching.
- **Dean’s List**, FAST NUCES: Fall 2023, Spring 2024, Fall 2024.
- **6th nationally**, SSC-II FBISE, 2021.
- **Runner-up**, AI Nexus Ignite Challenge, ACM-FAST Karachi, 2023 · **Runner-up**, GitWit Bugathon, 2025.